Generative AI for Software Developers

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Introduction

- Who you are
- What you’re working on
- Experience with
  - Python
  - Jupyter Notebooks
  - Neural networks
  - AI/ML software development
- Expectations

Course Objectives

- Learn how to use Ml and AI in a practical way to solve problems
  - Learn what ML and AI are and how they work
  - Develop an *intuitive* understanding of common learning algorithms and neural-network architectures
  - Learn what kinds of problems ML and AI can solve (and which ones they can't)
  - Learn how to use popular ML and AI frameworks to build sophisticated models
- Goal is to build ML/AI-savvy software developers, not data scientists!

Overview

- Introduction
  - AI/ML
  - Neural Networks
  - Backpropagation
- LLMs Part 1: Tokenization and Embeddings
- LLMs Part 2: Transformers
- Tools
- Retrieval Argumented Generation (RAG)
- Image Generation
- Agents

Logistics

- Class hours
  - 9 AM – 6 PM CST
- Courseware
  - https://github.com/ronaldsumida/GenerativeAI.git